

Channel relaxation and stationary-state occupation gaps

Asadullah Bhuiyan
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We distinguish the relaxation spectrum of an outcome-averaged fermionic channel from the occupation spectrum of its stationary correlation matrix. For a fixed sequence of perfect occupation resets, we derive the complete charge-neutral correlation hierarchy and show that the neutral and parity-even many-body channel gaps are exactly $1 - \rho(A)^2$, where A is the ordered product of complementary single-particle projectors. This equality requires more than closure of the two-point equation. Explicit local and nonlocal examples separate stationary purity gaps from dynamical gaps. We also establish the gap relations under probabilistic application, Poissonization, and finite-time Lindblad evolution, with their sector and locality qualifications. Verified hard-wall data illustrate the distinction for fixed-width and square geometries; finite-size occupation gaps are not interpreted as thermodynamic extrapolations.

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Appendix A: Elementary resets and covariance evolution

There are two different questions behind the word *gap*. One asks how rapidly a perturbation of a state decays under repeated evolution. The other asks whether the stationary correlation matrix has an occupation close to $1/2$. The first concerns the map that evolves a state; the second concerns the state reached by that map. We will derive both objects explicitly rather than infer one spectrum from the other.

The argument proceeds in three stages. First, we derive the action of one measurement-and-feedback step and compose the steps into a cycle. Second, we identify the covariance relaxation spectrum and prove when it determines the many-body gap. Third, we compare this dynamical information with stationary occupations and explain what changes on passing to continuous time. Simple matrix examples will make the distinctions concrete before we return to the numerical data.

We consider the physical-layer channel obtained after discarding the measurement record and fresh ancillary modes of the adaptive circuit of Bhuiyan, Pan, and Jian (BPJ) [1]. The assumptions throughout Apps. A and B are perfect feedback, normalized measured modes, and a fixed sequence repeated each cycle. The measured overcomplete Wannier (OW) modes need not be orthogonal. Support truncation is allowed, provided each retained mode is normalized. We retain the measurement dephasing already present in the reset channel; we do not append an additional dephasing operation.

1. Kraus representation and exact dissipator identity

Let $\{\hat{\psi}_a, \hat{\psi}_b^\dagger\} = \delta_{ab}$ for N fermion modes, and define

$$\hat{\chi}_j = \sum_a v_{ja} \hat{\psi}_a, \quad v_j^\dagger v_j = 1, \quad \hat{N}_j = \hat{\chi}_j^\dagger \hat{\chi}_j. \quad (\text{A1})$$

Here v_j is a column of coefficients: the definition of $\hat{\chi}_j$ contains no implicit complex conjugation. Emptying and filling this mode correspond, respectively, to

$$\begin{aligned}\tilde{\mathcal{E}}_{j,0}(\hat{\rho}) &= \hat{\chi}_j \hat{\rho} \hat{\chi}_j^\dagger + (\hat{\mathbf{1}} - \hat{N}_j) \hat{\rho} (\hat{\mathbf{1}} - \hat{N}_j), \\ \tilde{\mathcal{E}}_{j,1}(\hat{\rho}) &= \hat{\chi}_j^\dagger \hat{\rho} \hat{\chi}_j + \hat{N}_j \hat{\rho} \hat{N}_j.\end{aligned}\tag{A2}$$

Since $\hat{N}_j^2 = \hat{N}_j$ and $\hat{\chi}_j \hat{\chi}_j^\dagger = \hat{\mathbf{1}} - \hat{N}_j$, each pair of Kraus operators satisfies $\sum_m \hat{K}_m^\dagger \hat{K}_m = \hat{\mathbf{1}}$. These are completely positive trace-preserving maps.

The meaning of a reset can be seen without any lattice structure. For a single mode, in the empty/occupied basis, write

$$\hat{\rho} = \begin{pmatrix} 1-f & z \\ z^* & f \end{pmatrix}, \quad \tilde{\mathcal{E}}_0(\hat{\rho}) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \tilde{\mathcal{E}}_1(\hat{\rho}) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.\tag{A3}$$

Here f is the initial occupation and z is a coherence; their allowed values are fixed by positivity of $\hat{\rho}$. The reset forgets both. A physical parity-even single-mode state has $z = 0$; retaining z here also displays the action on the formal odd sector. In the emptying map, an occupied input loses its particle through $\hat{\chi}$, while an empty input is left empty by $\hat{\mathbf{1}} - \hat{N}$. The filling map exchanges the two roles. Summing the branches discards the record, so the two alternatives are added as density matrices, not as coherent amplitudes.

With the normalization

$$\mathcal{D}[\hat{L}](\hat{\rho}) = \hat{L} \hat{\rho} \hat{L}^\dagger - \frac{1}{2} \{ \hat{L}^\dagger \hat{L}, \hat{\rho} \},\tag{A4}$$

expanding the projector terms in Eq. (A2) gives

$$\begin{aligned}\tilde{\mathcal{E}}_{j,0} &= \text{Id} + \mathcal{D}[\hat{\chi}_j] + \mathcal{D}[\hat{N}_j], \\ \tilde{\mathcal{E}}_{j,1} &= \text{Id} + \mathcal{D}[\hat{\chi}_j^\dagger] + \mathcal{D}[\hat{N}_j].\end{aligned}\tag{A5}$$

For example, the two anticommutators in the emptying case sum to $-\{\hat{N}_j, \hat{\rho}\}$, while the remaining terms are $\hat{\chi}_j \hat{\rho} \hat{\chi}_j^\dagger + \hat{N}_j \hat{\rho} \hat{N}_j$. Adding $\hat{\rho}$ reproduces the first line of Eq. (A2). The filling case follows using $\hat{\chi}_j \hat{\chi}_j^\dagger = \hat{\mathbf{1}} - \hat{N}_j$. Equation (A5) is an exact discrete-channel identity, not a short-time approximation or a claim that $\tilde{\mathcal{E}}_j = e^{\mathcal{L}_j}$.

2. Closed two-point equation

For the outcome-averaged state define

$$\overline{G}_{ab} = \text{Tr}(\hat{\rho} \hat{\psi}_a^\dagger \hat{\psi}_b), \quad P_j = v_j v_j^\dagger, \quad Q_j = \mathbf{1} - P_j.\tag{A6}$$

The total charge is $\langle \hat{N}_F \rangle = \text{tr} \overline{G}$, with tr the single-particle trace and Tr the many-body trace. No Gaussian-state assumption enters this definition. The canonical anticommutation relations give

$$[\hat{N}_j, \hat{\psi}_b] = -v_{jb}^* \hat{\chi}_j, \quad [\hat{N}_j, \hat{\psi}_a^\dagger] = v_{ja} \hat{\chi}_j^\dagger.\tag{A7}$$

Using these identities in $\mathcal{D}^\dagger[\hat{N}_j](\hat{O}) = -\frac{1}{2}[\hat{N}_j, [\hat{N}_j, \hat{O}]]$, and evaluating the gain/loss term on $\hat{O} = \hat{\psi}_a^\dagger \hat{\psi}_b$, gives the two contributions

$$\begin{aligned}(\delta \overline{G})_{\text{gain/loss}} &= -\frac{1}{2}(P_j \overline{G} + \overline{G} P_j) + s_j P_j, \\ (\delta \overline{G})_{\text{deph}} &= P_j \overline{G} P_j - \frac{1}{2}(P_j \overline{G} + \overline{G} P_j).\end{aligned}\tag{A8}$$

Thus the exact elementary update is

$$\overline{G}' = Q_j \overline{G} Q_j + s_j P_j, \quad s_j \in \{0, 1\}.\tag{A9}$$

In a basis beginning with the measured mode, this erases its cross correlations and sets its occupation to s_j , leaving the orthogonal block unchanged. Explicitly, in that basis,

$$\overline{G} = \begin{pmatrix} f & b^\dagger \\ b & G_\perp \end{pmatrix} \quad \mapsto \quad \overline{G}' = \begin{pmatrix} s_j & 0 \\ 0 & G_\perp \end{pmatrix}.\tag{A10}$$

The complementary projector Q_j therefore retains only the part of the old correlations orthogonal to the measured mode, while $s_j P_j$ inserts the new occupation. Equations (A1) and (A6) fix the conjugation convention needed to use $P_j = v_j v_j^\dagger$.

One cycle is the ordered composition

$$\tilde{\mathcal{E}} = \prod_{j=M}^1 \tilde{\mathcal{E}}_{j,s_j} \equiv \tilde{\mathcal{E}}_{M,s_M} \circ \cdots \circ \tilde{\mathcal{E}}_{1,s_1}, \quad (\text{A11})$$

where step 1 acts first. Iteration of Eq. (A9) yields

$$\begin{aligned} \overline{G}' &= A \overline{G} A^\dagger + B, \\ A &= Q_M \cdots Q_1, \quad B = \sum_{j=1}^M s_j R_j P_j R_j^\dagger, \\ R_j &= Q_M \cdots Q_{j+1}, \quad R_M = \mathbf{1}. \end{aligned} \quad (\text{A12})$$

Noncommuting projectors make the order physical. The targets s_j enter B , whereas A depends only on the measured modes and order. The word *affine* refers to the state-independent source B : the update is a linear transformation plus an additive term. After n complete cycles, it reads

$$\overline{G}_n = A^n \overline{G}_0 (A^\dagger)^n + \sum_{\ell=0}^{n-1} A^\ell B (A^\dagger)^\ell. \quad (\text{A13})$$

The first term remembers the initial state; the sum collects occupations injected during earlier cycles and subsequently propagated by the remaining resets. In particular, the stationary equation is affine:

$$G_{\text{ss}} = A G_{\text{ss}} A^\dagger + B. \quad (\text{A14})$$

For $r \equiv \rho(A) < 1$ its unique solution is $G_{\text{ss}} = \sum_{n=0}^{\infty} A^n B (A^\dagger)^n$. Stationarity therefore does not require an eigenvalue 1 of A . This point is essential: the many-body channel has a stationary eigenvector, but the smaller matrix A describes the survival of initial correlations, not the entire state including its source.

Appendix B: Relaxation spectrum and many-body gap

1. Definitions and spectral inclusion

An eigenoperator $\hat{X}$ of the channel satisfies $\tilde{\mathcal{E}}(\hat{X}) = \mu \hat{X}$. After n cycles its amplitude is multiplied by μ^n . Thus $|\mu|$ determines decay and the phase of μ determines oscillation. A multiplier 0.9 decays more slowly than 0.2; a multiplier -0.9 has the same decay envelope as 0.9 but alternates in sign. The slowest relaxing mode is therefore the one with modulus closest to 1, not the smallest eigenvalue.

Fix an invariant linear operator sector $\mathcal{V}$ containing a stationary density matrix. The channel preserves the trace-zero subspace $\mathcal{V}_0$. This is the natural space of perturbations: the difference of any two normalized density matrices has zero trace. In particular, writing $\hat{\rho} = \hat{\rho}_{\text{ss}} + \delta\hat{\rho}$ gives

$$\tilde{\mathcal{E}}^n(\hat{\rho}) = \hat{\rho}_{\text{ss}} + \tilde{\mathcal{E}}^n(\delta\hat{\rho}), \quad \text{Tr}(\delta\hat{\rho}) = 0. \quad (\text{B1})$$

The first term is supposed to remain; only the second must decay for relaxation to occur. Excluding the stationary eigenvalue does not hide a slow perturbation: it removes the reference state itself. Define the absolute relaxation gap and logarithmic rate by

$$\begin{aligned} r_{\mathcal{V}} &= \rho(\tilde{\mathcal{E}}|_{\mathcal{V}_0}), & g_{\mathcal{V}} &= 1 - r_{\mathcal{V}}, \\ \kappa_{\mathcal{V}} &= -\log r_{\mathcal{V}}. \end{aligned} \quad (\text{B2})$$

When $r_{\mathcal{V}} < 1$, the stationary eigenvalue is simple in this sector and all other eigenvalues lie strictly inside the unit circle; the channel is mixing [2, 3]. Removing the stationary mode means restricting away from this one fixed direction, not discarding all troublesome eigenvalues. An additional stationary state supplies a trace-zero fixed direction, and

any other unit-modulus eigenvalue also gives $g_V = 0$. One can instead define a gap above an entire fixed-point space, but that is not a claim of relaxation to a unique state.

The two gap conventions encode the same multiplier in different units. If $r_V = 1 - g_V$, the slow envelope is $r_V^n = e^{-n\kappa_V}$, and

$$\kappa_V = -\log(1 - g_V) = g_V + \frac{1}{2}g_V^2 + \dots \quad (\text{B3})$$

The dimensionless channel gap and the decay rate per cycle agree only near a closing. For a large gap, replacing one by the other can substantially change the numerical value.

The absolute gap is $1 - \max |\mu|$, not $\min |1 - \mu|$. For a Lindbladian $\mathcal{L}$, the corresponding decay rate is

$$\delta_{\mathcal{L}, V} = - \max_{\lambda \in \text{spec}(\mathcal{L}|_{V_0})} \text{Re } \lambda. \quad (\text{B4})$$

The stationary zero is excluded under the same sector convention. Additional zeros or purely imaginary modes obstruct mixing.

There are three matrix sizes to distinguish. For N fermion modes, A and $\overline{G}$ are $N \times N$ matrices, a general linear map on their entries is $N^2 \times N^2$, and the unrestricted many-body channel acts on an operator space of dimension 4^N . The density matrix itself is $2^N \times 2^N$. The simplification is that a tractable part of the exponentially larger evolution closes on the two-point function.

Subtracting Eq. (A14) gives

$$\delta G' = A \delta G A^\dagger, \quad \text{vec}(\delta G') = (A^* \otimes A) \text{vec}(\delta G). \quad (\text{B5})$$

Column vectorization means stacking the columns of δG into one vector. It is a bookkeeping device, not another physical approximation. To see the spectrum directly, suppose first that $A w_i = a_i w_i$ gives an eigenbasis. Then

$$A(w_i w_j^\dagger) A^\dagger = a_i a_j^* w_i w_j^\dagger. \quad (\text{B6})$$

Each side of the covariance contributes one factor. Schur triangularization gives the same eigenvalues even without a complete eigenbasis. The covariance multipliers are therefore $a_i a_j^*$, including the positive multiplier $|a_i|^2$. Consequently

$$g_C = 1 - r^2, \quad \kappa_C = -2 \log r, \quad r = \max_i |a_i|. \quad (\text{B7})$$

No eigenvalue of A is removed in computing r . The square comes from the two-sided action, not from a choice of gap convention. The spectral radius, rather than the largest singular value, controls asymptotic decay. Below we take $0 < r < 1$; if $r = 0$, the finite-dimensional homogeneous map is nilpotent and its logarithmic rate is infinite.

2. A two-mode example of the projector product

For a concrete illustration, empty two real normalized modes in sequence, with coefficient vectors $v_1 = (1, 0)^\top$ and $v_2 = (\cos \theta, \sin \theta)^\top$. Writing $c_\theta = \cos \theta$ and $s_\theta = \sin \theta$, their complementary projectors and ordered product are

$$Q_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad Q_2 = \begin{pmatrix} s_\theta^2 & -c_\theta s_\theta \\ -c_\theta s_\theta & c_\theta^2 \end{pmatrix}, \quad A = Q_2 Q_1 = \begin{pmatrix} 0 & -c_\theta s_\theta \\ 0 & c_\theta^2 \end{pmatrix}. \quad (\text{B8})$$

The eigenvalues of A are $0, c_\theta^2$, whereas the covariance map has eigenvalues $0, 0, 0, c_\theta^4$. Hence $r = c_\theta^2$ and $g_C = 1 - c_\theta^4$. Nearly parallel measured modes leave an almost unmeasured direction and give slow relaxation. Orthogonal measured modes erase all initial correlations in one cycle. At $\theta = 0$ the second physical mode is never measured, and the gap vanishes.

Both targets in this example are empty, so $B = 0$. For $0 < \theta < \pi/2$ the stationary matrix is $G_{\text{ss}} = 0$ even though the relaxation rate varies with θ . The example also separates eigenvalues from singular values: the largest singular value of A is $|c_\theta|$, not c_θ^2 . A largest-singular-value calculation would therefore not give the same asymptotic rate.

3. Why covariance closure initially gives only a bound

Covariance closure alone proves an upper bound. The adjoint channel preserves the span of $\hat{\mathbf{1}}$ and the bilinears $\hat{\psi}_a^\dagger \hat{\psi}_b$. After centering about G_{ss} , the covariance eigenvalues are contained in the many-body spectrum. Thus, for any mixing sector containing these operators,

$$r_V \geq r^2, \quad g_V \leq g_C, \quad \kappa_V \leq \kappa_C. \quad (\text{B9})$$

A positive upper bound cannot rule out a vanishing many-body gap. Knowing one family of decaying observables need not identify the slowest observable in the entire system. An unseen higher-order correlation could decay more slowly. To replace the inequality by an equality, we must rule out precisely that possibility. The continuous-time correlation hierarchy of Barthel and Zhang [4], especially their Proposition 7, provides the analogous spectral-inclusion reasoning. Equality here will follow from the specific reset structure, not from closure by itself.

The covariance mode can also be exhibited as an observable. For $A^\dagger u = a^* u$, define

$$\hat{O}_u = \sum_{ab} u_a^* u_b \left[\hat{\psi}_a^\dagger \hat{\psi}_b - (G_{ss})_{ab} \hat{\mathbf{1}} \right]. \quad (\text{B10})$$

Since its expectation is $u^\dagger \delta G u$, $\tilde{\mathcal{E}}^\dagger(\hat{O}_u) = |a|^2 \hat{O}_u$. Choosing $|a| = r$ realizes the slow neutral observable. A right eigenvector $Aw = aw$ instead gives the right covariance eigenmatrix $ww^\dagger$; these left and right constructions should not be interchanged for nonnormal A .

4. Higher moments and exterior powers

Within the parity-even sector, a perfect reset never creates a higher-degree observable from a lower-degree one. We use *degree* to count creation and annihilation operators in a normal-ordered monomial. Bilinears have degree two; the corresponding four-point moments have degree four. We keep the full moments, not a Gaussian factorization of them.

Choose a single-particle basis beginning with $\hat{\chi}_j$, and let $\hat{B}$ contain only orthogonal modes. On even total operators, direct substitution into Eq. (A2) gives

$$\begin{aligned} \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{B}) &= \hat{B} & (\hat{B} \text{ even}), \\ \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{\chi}_j^{(\dagger)} \hat{B}) &= 0 & (\hat{B} \text{ odd}), \\ \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{N}_j \hat{B}) &= s \hat{B} & (\hat{B} \text{ even}). \end{aligned} \quad (\text{B11})$$

A single measured leg is erased, while a measured occupation pair is replaced by its target and lowers the degree by two. This is an operator identity and makes no use of Wick factorization.

For increasing p -element index sets I, J , introduce

$$F_{I;J}^{(p)} = \text{Tr} \left(\hat{\rho} \hat{\psi}_{i_1}^\dagger \cdots \hat{\psi}_{i_p}^\dagger \hat{\psi}_{j_p} \cdots \hat{\psi}_{j_1} \right). \quad (\text{B12})$$

For $p = 1$ these are the entries of $\bar{G}$; for $p = 2$ they are four-point moments. In vector form, the first levels of their update have the schematic structure

$$\begin{pmatrix} 1 \\ F^{(1)} \\ F^{(2)} \\ \vdots \end{pmatrix}' = \begin{pmatrix} 1 & 0 & 0 & \cdots \\ b_1 & H_1 & 0 & \cdots \\ b_2 & T_{21} & H_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} 1 \\ F^{(1)} \\ F^{(2)} \\ \vdots \end{pmatrix}. \quad (\text{B13})$$

Here b_p and T_{21} denote lower-order source terms; their values are not needed to determine the eigenvalues of this block-triangular matrix. What matters is the diagonal block H_p propagating the degree- $2p$ part.

The homogeneous degree- $2p$ term is obtained by projecting each leg onto the orthogonal complement of the measured mode. Antisymmetrization turns the p creation-leg factors into the exterior power $\bigwedge^p Q_j$ and the annihilation-leg factors into its conjugate. The one-cycle homogeneous block is therefore

$$H_p[I, J; K, L] = \det A[I, K] \det A[J, L]^*. \quad (\text{B14})$$

The exterior-power notation is simply the antisymmetrized action on several fermionic indices. For example, on two creation legs,

$$\left(\bigwedge^2 A \right)_{\{i,j\}, \{k,\ell\}} = A_{ik} A_{j\ell} - A_{i\ell} A_{jk}. \quad (\text{B15})$$

The minus sign is the fermionic exchange sign. For p legs, the same sum over signed permutations is the $p \times p$ determinant in Eq. (B14). Indeed, the Cauchy–Binet identity composes the minors of successive Q_j in exactly the

order in Eq. (A12). All source terms from Eq. (B11) belong to lower-degree blocks. The full hierarchy, ordered by p , is block triangular.

The eigenvalues of $\bigwedge^p A$ are the products of p distinct eigenvalues of A . This remains true for defective A , as follows by taking exterior powers of its Schur-triangular form. Hence the neutral-sector spectrum, with algebraic multiplicities, is

$$\Lambda_{I,J} = \prod_{i \in I} a_i \prod_{j \in J} a_j^*, \quad |I| = |J| = p, \quad 0 \leq p \leq N. \quad (\text{B16})$$

The sets I and J are independently chosen and may overlap. In particular, the same eigenvalue may appear on both legs.

To see that no neutral operators are omitted, the normal-ordered monomials with equal numbers of creation and annihilation operators form a basis. One may construct each number-preserving Fock-space matrix unit by multiplying its creation and annihilation string by occupation or vacancy projectors on the remaining sites. Expanding these projectors gives precisely such monomials. Their number is

$$\sum_{p=0}^N \binom{N}{p}^2 = \binom{2N}{N}, \quad (\text{B17})$$

which is also the dimension of all operators block diagonal in particle number. Spanning and this dimension count establish completeness of Eq. (B16).

The constant block $p = 0$ contributes one eigenvalue 1. For $p \geq 1$, $|\Lambda_{I,J}| \leq r^{2p} \leq r^2$, and the bound is attained at $p = 1$ with $I = J$ corresponding to $|a_i| = r$. Thus the stationary eigenvalue is simple and

$$\boxed{g_{q=0} = g_C = 1 - r^2, \quad \kappa_{q=0} = \kappa_C.} \quad (\text{B18})$$

Higher neutral correlations cannot have a slower spectral decay.

5. Neutral, even, and odd sectors

Charge neutrality means $[\hat{N}_F, \rho] = 0$, with $\hat{N}_F = \sum_a \hat{\psi}_a^\dagger \hat{\psi}_a$. It permits mixtures of different particle numbers. The reset channel preserves neutrality, but it does not conserve particle number: its Kraus operators have definite, not necessarily zero, charge. Both maximally mixed initial states and fixed-number Slater states belong to the neutral sector.

Examples clarify the distinction between charge and parity. $\hat{\psi}_1^\dagger \hat{\psi}_2$ is neutral because it has one creation and one annihilation operator. $\hat{\psi}_1 \hat{\psi}_2$ is not neutral, but it is parity even because it contains two fermionic operators. $\hat{\psi}_1$ is parity odd. An invariant operator sector groups quantities whose evolution stays within that group; it is not necessarily a fixed-particle-number subspace of state vectors.

For a monomial with p creation and ℓ annihilation operators, the same even-operator argument gives a homogeneous block $\bigwedge^p A \otimes (\bigwedge^\ell A)^*$, up to the chosen vectorization order. Every nonconstant even monomial has $p + \ell \geq 2$. Its eigenvalue modulus is at most $r^{p+\ell} \leq r^2$, and the neutral bilinears attain r^2 . Consequently

$$g_{\text{even}} = g_{q=0} = g_C, \quad \kappa_{\text{even}} = \kappa_C. \quad (\text{B19})$$

For completeness, the specified Kraus maps also define an extension to parity-odd operators. Let $\hat{P}_F = (-1)^{\hat{N}_F}$ and $\mathcal{S}(\hat{O}) = \hat{P}_F \hat{O}$, so $\mathcal{S}^2 = \text{Id}$. Writing $\hat{J} = \hat{\chi}$ or $\hat{\chi}^\dagger$ for the odd Kraus operator and $\hat{R}$ for the even projector Kraus operator,

$$(\mathcal{S} \tilde{\mathcal{E}}_j^\dagger \mathcal{S})(\hat{O}) = -\hat{J}^\dagger \hat{O} \hat{J} + \hat{R} \hat{O} \hat{R}. \quad (\text{B20})$$

On odd operators, this extra minus sign cancels the sign from moving an odd orthogonal-mode string past $\hat{J}$. The transformed map therefore leaves an orthogonal odd string unchanged, erases a single measured leg, and sends $\hat{N} \hat{B}$ to $s \hat{B}$. Its homogeneous hierarchy has odd $p + \ell$, with maximal modulus r attained by one-leg blocks. Similarity preserves the spectrum and respects composition across the cycle. On unrestricted operator space,

$$g_{\text{full}} = 1 - r, \quad \kappa_{\text{full}} = -\log r = \kappa_C/2. \quad (\text{B21})$$

Physical fermionic density matrices are parity even; Eq. (B19) is the relevant many-body gap for the preparations considered here. Equation (B21) explains why an unrestricted spectral definition can differ without contradicting covariance closure.

These equalities are specific to the exact reset hierarchy. Imperfect feedback, extra dephasing, or averaging over a new random schedule each cycle requires reexamining the higher moments. Nonnormality can also produce long transients and Jordan-polynomial prefactors. A spectral gap fixes the asymptotic exponential envelope at fixed size, not a size-uniform mixing time or a conditioned trajectory-purification time.

Appendix C: Stationary occupation gaps and counterexamples

1. Distinct spectra

Let $n_a \in [0, 1]$ be the eigenvalues of the full-system stationary correlation matrix G_{ss} . We use

$$\Delta = \min_a |n_a - \tfrac{1}{2}|, \quad \Delta_{\text{pur}} = 2\Delta = \min_a |1 - 2n_a|. \quad (\text{C1})$$

These are stationary occupation diagnostics; they are not eigenvalue gaps of a channel. Our normalization makes the purity gap unity for a pure number-conserving Gaussian state and zero when an occupation equals 1/2. Other normalizations, including squared conventions, must be converted before comparison. The full-system matrix is also distinct from a subsystem restriction used for an entanglement spectrum.

The reference point 1/2 comes from the occupation of a maximally mixed fermion, not from a stationary eigenvalue of the evolution. For example, occupations 0.01 and 0.99 are far from maximal mixing and give $\Delta_{\text{pur}} = 0.98$, while an occupation 0.499 gives $\Delta_{\text{pur}} = 0.002$. Neither statement says how long it took to prepare those occupations. Also, the symbol $\rho(A)$ denotes a *spectral radius*; it is not a density matrix $\hat{\rho}$. Likewise, n_a labels occupations of G_{ss} , whereas μ labels multipliers of a dynamical map.

For a Gaussian state the single-particle modular energies are

$$\epsilon_a = \log \frac{1 - n_a}{n_a}. \quad (\text{C2})$$

For a non-Gaussian outcome-averaged state, this defines the Gaussian reference state sharing its two-point function, not the spectrum of its actual many-body modular Hamiltonian $-\log \hat{\rho}_{ss}$. An occupation at 1/2 denotes maximal mixing of that covariance mode; it does not identify a conserved observable or a dark mode of the evolution.

The separation is already visible in Eq. (A12): relaxation of δG depends on A , while G_{ss} also depends on B . For gauge-invariant quasi-free channels, complete positivity in this convention requires $0 \leq B \leq \mathbf{1} - AA^\dagger$ [5]. For example, $A = \sqrt{q} \mathbf{1}$ and $B = (1 - q)G_0$, with $0 \leq G_0 \leq \mathbf{1}$ and $0 \leq q < 1$, give $G_{ss} = G_0$ and $g_C = 1 - q$, independently of the occupation gap of G_0 . For a single occupation, the same construction is the elementary recursion

$$n_{m+1} = qn_m + (1 - q)n_*, \quad n_m - n_* = q^m(n_0 - n_*). \quad (\text{C3})$$

The number n_* specifies where the evolution ends; q specifies how much of the deviation survives each step. One may change n_* from nearly empty to exactly half-filled without changing q . This is the simplest way to see why a purity-gap closing does not force slow relaxation.

This Gaussian-channel construction is illustrative; closure of the two-point equation does not itself make our full reset channel Gaussian on arbitrary inputs.

2. Global and local mixing examples

For any density matrix $\hat{\sigma}$, let $\mathcal{T}_\sigma(\hat{X}) = \text{Tr}(\hat{X})\hat{\sigma}$. The replacement channel

$$\tilde{\mathcal{E}}_p = (1 - p)\text{Id} + p\mathcal{T}_\sigma, \quad 0 < p \leq 1, \quad (\text{C4})$$

has the unique stationary state $\hat{\sigma}$ and multiplier $1 - p$ on every trace-zero operator. Its gap is p , independently of the stationary spectrum. To verify the claim, write $\hat{\rho} = \hat{\sigma} + \delta\hat{\rho}$ with $\text{Tr} \delta\hat{\rho} = 0$. Since $\mathcal{T}_\sigma(\delta\hat{\rho}) = 0$, the channel maps the perturbation to $(1 - p)\delta\hat{\rho}$ while leaving $\hat{\sigma}$ unchanged. There is no additional slow subspace to inspect. Taking $\hat{\sigma} = \hat{\mathbf{1}}/d$ gives a maximally mixed endpoint and, for a fermionic Fock space, $G_{ss} = \mathbf{1}/2$. The generator $\gamma(\mathcal{T}_\sigma - \text{Id})$ has nonstationary eigenvalues $-\gamma$, and

$$e^{t\gamma(\mathcal{T}_\sigma - \text{Id})} = \mathcal{T}_\sigma + e^{-\gamma t}(\text{Id} - \mathcal{T}_\sigma). \quad (\text{C5})$$

Global replacement is generally nonlocal, so it alone does not address locality constraints.

A strictly local example is independent depolarization of spins. Let $\mathcal{T}_i$ replace spin i by $\mathbf{1}_2/2$ while tracing out its old state. Then

$$\mathcal{L}_{\text{loc}} = \gamma \sum_i (\mathcal{T}_i - \text{Id}) = \frac{\gamma}{4} \sum_{i,a=x,y,z} \mathcal{D}[\hat{\sigma}_i^a]. \quad (\text{C6})$$

A Pauli string of weight w has eigenvalue $-\gamma w$. The weight counts sites with a nonidentity Pauli factor. Replacing any such site erases that string, so each of its w sites contributes one decay rate γ . The identity string has weight zero and is stationary; the slowest nonconstant strings have weight one. The unique stationary state is maximally mixed and the gap is γ , uniformly in the number of spins. Sampling this local semigroup at time τ gives a channel gap $1 - e^{-\gamma\tau}$. No locality-preserving spin-to-fermion mapping is needed or asserted.

For a single fermionic mode, balanced gain and loss give

$$\mathcal{L}_{\text{bal}} = \eta(\mathcal{D}[\hat{c}] + \mathcal{D}[\hat{c}^\dagger]), \quad \dot{n} = \eta(1 - 2n). \quad (\text{C7})$$

Its full spectrum is $\{0, -\eta, -\eta, -2\eta\}$ and its stationary state is $\hat{\mathbf{1}}/2$. The even population sector decays at 2η ; the unrestricted gap is η . Thus even the simplest fermionic example requires stating which operator sector is meant.

3. Period-two channel and stationary-mode exclusion

The channel

$$\tilde{\mathcal{E}}_{\text{flip}}(\hat{\rho}) = \hat{c}\hat{\rho}\hat{c}^\dagger + \hat{c}^\dagger\hat{\rho}\hat{c} \quad (\text{C8})$$

is trace preserving because $\hat{c}^\dagger\hat{c} + \hat{c}\hat{c}^\dagger = \hat{\mathbf{1}}$. Its action on an arbitrary matrix is

$$\tilde{\mathcal{E}}_{\text{flip}} \begin{pmatrix} x & z \\ w & y \end{pmatrix} = \begin{pmatrix} y & 0 \\ 0 & x \end{pmatrix}. \quad (\text{C9})$$

The diagonal entries are exchanged, while off-diagonal entries are erased. A fixed matrix must have $x = y$ and $z = w = 0$, proving directly that the stationary space is spanned only by $\hat{\mathbf{1}}$. On $\hat{\mathbf{1}}, \hat{\sigma}_z, \hat{\sigma}_x, \hat{\sigma}_y$ it has eigenvalues $1, -1, 0, 0$, respectively. Its stationary space is one dimensional, spanned by $\hat{\mathbf{1}}$, yet it is not mixing: a population imbalance changes sign every step. The absolute channel gap is zero because of -1 . The two zero multipliers instead describe coherences erased in one step. Neither taking the smallest eigenvalue nor simply deleting all unit-modulus eigenvalues gives the relaxation gap.

Its Poissonized generator $\gamma(\tilde{\mathcal{E}}_{\text{flip}} - \text{Id})$ has spectrum $\{0, -2\gamma, -\gamma, -\gamma\}$ and is precisely Eq. (C7) with $\eta = \gamma$. Randomizing the number of flips removes the period-two obstruction without changing the stationary state. This supplies a counterexample to a general reverse implication from a gapped Poissonized generator to a mixing discrete channel.

Finally, a gapped Hamiltonian can have a maximally mixed infinite-temperature Gibbs state: for example $\hat{H} = \varepsilon \sum_i \hat{n}_i$ has excitation gap ε but $\hat{\rho}_{\beta=0} = \hat{\mathbf{1}}/2^N$. This only distinguishes a Hamiltonian excitation gap from a state's occupation gap; unitary evolution under that Hamiltonian does not prepare this state as a unique attractor.

Appendix D: Continuous-time connection

1. Probabilistic application and Poissonization

For an arbitrary channel $\tilde{\mathcal{E}}$, define

$$\tilde{\mathcal{E}}_p = (1 - p)\text{Id} + p\tilde{\mathcal{E}}, \quad 0 < p \leq 1. \quad (\text{D1})$$

Its fixed-point space is exactly that of $\tilde{\mathcal{E}}$, since $\tilde{\mathcal{E}}_p - \text{Id} = p(\tilde{\mathcal{E}} - \text{Id})$. Each eigenvalue transforms as $\mu_p = 1 - p + p\mu$, with no diagonalizability assumption. If $g = 1 - \max_{\nu_0} |\mu| > 0$, then

$$|\mu_p| \leq 1 - p + p|\mu| \leq 1 - pg, \quad \boxed{g(p) \geq pg > 0}. \quad (\text{D2})$$

Hence the mixing gap cannot close at any fixed $p > 0$ along this particular deformation. The identity endpoint $p = 0$ is excluded: every state is then stationary. This is a spectral estimate, not an argument from continuity alone.

Taking $p = \gamma dt$ gives the generator

$$\mathcal{L}_{\text{cyc}} = \gamma(\tilde{\mathcal{E}} - \text{Id}) = \sum_a \mathcal{D}[\sqrt{\gamma} \hat{K}_a], \quad (\text{D3})$$

where $\hat{K}_a$ are Kraus operators for the whole channel and $\sum_a \hat{K}_a^\dagger \hat{K}_a = \hat{1}$. Equivalently, subtract the input state from the probabilistic update and divide by the elapsed time:

$$\frac{\hat{\rho}(t+dt) - \hat{\rho}(t)}{dt} = \gamma[\tilde{\mathcal{E}}(\hat{\rho}(t)) - \hat{\rho}(t)]. \quad (\text{D4})$$

The limit of repeated small steps is the continuous-time semigroup generated by this right-hand side. The Kraus expression in Eq. (D3) independently verifies that this is a valid Lindblad generator, not just a formal differential equation. Its semigroup has the explicit completely positive representation

$$e^{t\mathcal{L}_{\text{cyc}}} = e^{-\gamma t} \sum_{n=0}^{\infty} \frac{(\gamma t)^n}{n!} \tilde{\mathcal{E}}^n. \quad (\text{D5})$$

It applies a Poisson-distributed number of complete cycles and has the same stationary space as $\tilde{\mathcal{E}}$. Physically, the deterministic sequence inside a cycle is unchanged; only the times at which complete cycles occur are randomized. A mode with multiplier μ therefore acquires the averaged factor

$$e^{-\gamma t} \sum_{n=0}^{\infty} \frac{(\gamma t)^n}{n!} \mu^n = e^{\gamma t(\mu-1)}. \quad (\text{D6})$$

This also explains why phase oscillations can become damping: averaging over different numbers of steps can cancel them. The eigenvalues and gap obey

$$\begin{aligned} \lambda &= \gamma(\mu - 1), \\ \delta_{\mathcal{L}_{\text{cyc}}, \nu} &= \gamma \min_{\nu_0} (1 - \text{Re } \mu) \geq \gamma g. \end{aligned} \quad (\text{D7})$$

The converse fails: Eq. (C8) has $g = 0$ but a positive Poissonized decay rate. More generally, eigenvalues elsewhere on the unit circle have negative real parts after subtracting 1. For the lazy flip channel the nonstationary eigenvalues are $1 - 2p, 1 - p, 1 - p$; it mixes for $0 < p < 1$, although the original $p = 1$ channel does not.

For the reset covariance spectrum, the dominant multiplier r^2 is real, nonnegative, and present in the spectrum. It saturates both inequalities: its probabilistic multiplier is $1 - p + pr^2$, and its Poissonized exponent is $\gamma(r^2 - 1)$. All other covariance multipliers obey the previously established bounds, so none can be slower. This gives the exact identities

$$g_C(p) = p g_C, \quad \delta_{\mathcal{L}_{\text{cyc}}, C} = \gamma g_C. \quad (\text{D8})$$

By App. B, the same equalities hold in the neutral and even many-body sectors. The continuous-time rate in Eq. (D8) is γg_C , not $\gamma \kappa_C$. A per-step gap proportional to dt tends to zero when the clock step shrinks without implying slow relaxation in physical time. For example, applying a channel with probability γdt per tick means that most sufficiently short ticks do nothing. The number of opportunities per unit time grows as $1/dt$ and compensates the small gap per tick. Poissonization measures this physical-time decay; κ_C instead describes exactly one completed reset cycle per discrete step.

2. Finite-time semigroups and limits of the correspondence

If a relaxing Lindbladian is given first, its finite-time channel $\tilde{\mathcal{E}}_\tau = e^{\tau \mathcal{L}}$ has multipliers $e^{\tau \lambda}$. In the same invariant sector,

$$g(\tilde{\mathcal{E}}_\tau) = 1 - e^{-\tau \delta_{\mathcal{L}}}, \quad \kappa(\tilde{\mathcal{E}}_\tau) = \tau \delta_{\mathcal{L}}. \quad (\text{D9})$$

This spectral mapping statement differs from Poissonizing an arbitrary channel. In general $\tilde{\mathcal{E}}$ need not admit a Lindbladian logarithm; semigroup-embedding restrictions already occur for qubit channels [6]. Equation (D3) always constructs a generator, but its exponential is not generally $\tilde{\mathcal{E}}$.

Three qualifications are important. First, whole-cycle Kraus operators can extend across the system. A local gate sequence does not automatically make $\mathcal{L}_{\text{cyc}}$ a finite-range Lindbladian, especially when the sequential depth grows with size. Weakening each elementary reset separately instead produces, to first order, $\sum_j \gamma_j (\tilde{\mathcal{E}}_j - \text{Id})$. This generator need not share the stationary state of the ordered full-reset product. They do share a state if every reset fixes it individually, but such a common fixed state is an additional assumption.

Second, finite-dimensional eigenvalue identities do not bound transient amplification or all mixing prefactors. The projector product and its induced maps can be nonnormal, and defective blocks introduce powers of time multiplying exponentials. Third, thermodynamic gap preservation requires a size-independent lower bound on g , with p bounded away from zero or γ fixed. Positivity at each finite size is insufficient. Local mixing and clustering theorems such as those of Kastoryano and Eisert [7] have further regularity and reversibility or stronger mixing assumptions. They cannot be imported for the whole-cycle construction without checking those hypotheses.

Appendix E: Trajectory dynamics and relation to the literature

1. Conditioning versus discarding the record

For a complete measurement record $\mathbf{m}$ let $\hat{V}_{\mathbf{m}}$ be the corresponding Kraus product. Then

$$\begin{aligned} p_{\mathbf{m}} &= \text{Tr}(\hat{V}_{\mathbf{m}} \hat{\rho}_0 \hat{V}_{\mathbf{m}}^\dagger), & \hat{\rho}_{\mathbf{m}} &= \frac{\hat{V}_{\mathbf{m}} \hat{\rho}_0 \hat{V}_{\mathbf{m}}^\dagger}{p_{\mathbf{m}}}, \\ \bar{\rho}_n &= \sum_{\mathbf{m}} p_{\mathbf{m}} \hat{\rho}_{\mathbf{m}} = \tilde{\mathcal{E}}^n(\hat{\rho}_0). \end{aligned} \quad (\text{E1})$$

Linear observables obey $\sum_{\mathbf{m}} p_{\mathbf{m}} \text{Tr}(\hat{\rho}_{\mathbf{m}} \hat{O}) = \text{Tr}(\bar{\rho}_n \hat{O})$. Entropies do not: generally

$$\sum_{\mathbf{m}} p_{\mathbf{m}} S((\hat{\rho}_{\mathbf{m}})_R) \neq S((\bar{\rho}_n)_R), \quad (\text{E2})$$

where R is a spatial region. Different unravelings can yield the same averaged channel but different conditioned entanglement. The failure of entropy to commute with averaging is elementary. An equal ensemble of an empty and an occupied one-mode state has zero entropy in each member, but its averaged state $\hat{\mathbf{1}}/2$ has entropy $\log 2$. This is a statement about ensemble mixing, not an entanglement claim for that one mode; the same nonlinearity prevents replacing averaged subsystem entropies by the entropy of the averaged subsystem state. Accordingly, algebraic relaxation of trajectory-resolved entanglement is not, by itself, evidence that the one-copy outcome-averaged channel has zero gap. A relation would require an additional argument connecting that nonlinear record-dependent quantity to the relevant channel or a replicated evolution.

This distinction separates the conditioned purification discussed in BPJ Fig. 9 from the averaged correlation matrix and occupation spectrum in BPJ Fig. 7 [1]. Spatial algebraic correlations, temporal decay of averaged linear observables, and temporal decay of conditioned entanglement are three different diagnostics. Even a temporal power law over a finite window need not determine the asymptotic finite-size rate if transients or crossovers remain. None of these distinctions negates a measured power law; they identify which spectral claim that measurement can support.

2. Local dissipative examples and scope of existing results

A relevant finite-range example is Bardyn *et al.* [8], Eqs. (68)–(69) and Fig. 6. Their competing one-dimensional jump families are

$$\hat{L}_n^{(\ell)} = \frac{1}{2} [(\hat{a}_n^\dagger + \hat{a}_{n+\ell}^\dagger) + (\hat{a}_n - \hat{a}_{n+\ell})], \quad \ell = 1, 2. \quad (\text{E3})$$

With their normalized relative-rate convention, the bulk damping spectrum remains flat and gapped as the bulk purity gap closes at equal competing strengths. This is a finite-range real-space construction, not merely independent control of nonlocal momentum modes. The bulk qualifier is essential: open boundaries can carry zero-damping Majorana edge

modes. The example therefore does not establish a gapped full open-boundary generator. It establishes that locality alone does not equate the two bulk gaps.

Relations can reappear with extra structure. In the Gaussian setting, damping and noise jointly determine the stationary covariance; special dark-state or frustration-free conditions can lock these quantities [9]. Such restrictions are not a general identity between stationary purity and relaxation gaps. Likewise, the symmetry-class correspondence between quadratic Lindbladians and their unique Gaussian stationary states [10] is a statement about symmetry constraints, not equality of gaps. Its linear-jump assumptions do not automatically cover the number-dephasing term in Eq. (A5).

Discrete-channel studies also treat dynamical and invariant-state spectra as separate diagnostics. Spectral-support transitions in random Kraus maps [11] should not be conflated with a relaxation gap closing or a fermionic purity-gap crossing. Noisy Floquet channels provide examples with maximally mixed stationary states and nontrivial relaxation, while emphasizing the order of weak-noise and thermodynamic limits [12]. Broader random-channel results give further context for spectra and invariant states [13], not a universal relation between their gaps. For the present reset model, the precise many-body statement remains the hierarchy proof in App. B.

Appendix F: Numerical illustration and verification

1. Common protocol and postprocessing

We use two completed CPU campaigns of the canonical `run_markov_channel` implementation. Both use $\alpha_1 = 1, 3$, $\alpha_2 = 30$, $n_{\text{shell}} = 1$, hard-wall support truncation, all slabs active, periodic boundaries, zero twist, X trial orbitals, perfect correction, and double-precision complex arithmetic. Within each cell the order is $A+, A-, B+, B-$, with the raster- y schedule (x outer, y inner). The initial correlation matrix is $1/2$. These are exact outcome-averaged covariance calculations, not trajectory samples.

Each run contains 61 cycles, with saved occupation spectra at 0, 1, 5, 10, 20, 40, 60, 61. We report cycle 61. For all 24 runs, the last five successive Frobenius changes are below 10^{-10} , and the full and twirled purity gaps change by less than 2×10^{-10} between cycles 60 and 61. This is a numerical stationarity check, not a rigorous norm bound on the error to the fixed point of a nonnormal map.

The translation-twirled matrix is the postprocessed average

$$\mathcal{T}_y(G) = \frac{1}{N_y} \sum_{r=0}^{N_y-1} T_y^r G (T_y^\dagger)^r. \quad (\text{F1})$$

It is block diagonal in $k_y = 2\pi m/N_y$; its spectrum is the union of the spectra of those momentum blocks, not an average of the original eigenvalues. To see this, in a translation eigenbasis each matrix element between k_y and k'_y is multiplied by $N_y^{-1} \sum_r e^{ir(k_y - k'_y)} = \delta_{k_y, k'_y}$. Twirling removes couplings between distinct momenta but retains the matrix within each momentum block; those retained matrices must then be diagonalized. Twirling does not alter the simulated schedule or produce another dynamics run. In particular, $\mathcal{T}_y(G_{\text{ss}})$ need not itself be a fixed point of the raster-ordered channel.

2. Fixed-width systems

The fixed-width campaign uses $N_x = 20$ and $N_y = 20, 40, 60, 80, 100$, with inclusive walls at $x = 5, 15$. Table I reports the full and twirled purity gaps. The $\alpha_1 = 1$ values are small and nonmonotone; the $\alpha_1 = 3$ values remain near 0.986. Occupation half-distances Δ are one half of every tabulated entry.

Separate matched spectral calculations exist for these ten cases. Their covariance channel gaps decrease from 0.851086 to 0.835675 for $\alpha_1 = 1$, and from 0.961228 to 0.957611 for $\alpha_1 = 3$, between $N_y = 20$ and 100. These are g_C , not the logarithmic rates κ_C . Thus small stationary purity gaps coexist with large finite-size relaxation gaps in this fixed-width sequence. Increasing N_y at fixed N_x is not the two-dimensional fixed-aspect-ratio thermodynamic limit.

For a concrete reading of the numbers, at $(N_x, N_y, \alpha_1) = (20, 20, 1)$ the full purity gap 0.086790 means that the closest occupation is a distance 0.043395 from half filling. Independently, $g_C = 0.851086$ means that the slowest covariance multiplier is approximately 0.148914 per cycle. The first number locates an occupation in the final state; the second gives the asymptotic decay factor of a perturbation. Their sizes need not track one another.

The twirled spectra provide evidence of momentum-grid effects: the relevant branch at $\alpha_1 = 1$ comes closest to half filling near $|k_y| \simeq 0.05\pi$. Grids with $N_y = 40$ and 80 sample that region especially closely. The overlapping branch

TABLE I. Fixed width $N_x = 20$: stationary purity gaps $\Delta_{\text{pur}} = \min |1 - 2n|$. “Twirled” refers to Eq. (F1). All entries are regenerated from the saved cycle-61 occupation spectra, with six decimal places.

N_y	$\alpha_1 = 1$		$\alpha_1 = 3$	
	Full	Twirled	Full	Twirled
20	0.086790	0.092435	0.985595	0.985567
40	0.004357	0.003950	0.985659	0.985703
60	0.021813	0.022742	0.985667	0.985748
80	0.005699	0.005474	0.985668	0.985771
100	0.009523	0.009948	0.985668	0.985784

TABLE II. Square systems $N_x = N_y = L$: full and postprocessed twirled purity gaps, with the same definitions and precision as Table I. These are stationary occupation results, not a square-system channel-gap sweep.

L	$\alpha_1 = 1$		$\alpha_1 = 3$	
	Full	Twirled	Full	Twirled
20	0.086790	0.092435	0.985595	0.985567
30	0.024280	0.026040	0.985247	0.985250
40	0.004357	0.003950	0.985086	0.985106
50	0.022346	0.022455	0.984999	0.985031
60	0.021813	0.022741	0.984946	0.984988
70	0.005190	0.005699	0.984912	0.984961
80	0.005699	0.005474	0.984889	0.984943

shapes and nonmonotone minima are consistent with commensurability, but do not prove an exact crossing between grid points or a thermodynamic purity-gap closing. No interpolation or fit is used here.

3. Square systems and provenance

The square campaign uses $N_x = N_y = L = 20, 30, 40, 50, 60, 70, 80$. The inclusive wall positions are $\lfloor L/4 \rfloor$ and $\lfloor 3L/4 \rfloor$; their integer rounding is retained. Table II again shows small, nonmonotone $\alpha_1 = 1$ purity gaps and large $\alpha_1 = 3$ gaps. These data vary both dimensions and must not be pooled with the fixed-width sequence as though the geometries were identical. This new square *dynamics* sweep did not compute channel gaps. Table II makes no claim about their size dependence.

The accompanying build script verifies byte counts and SHA-256 digests for all 24 result/completion pairs, and independently recomputes both purity-gap columns from saved occupation spectra. It also verifies the ten matched fixed-width spectral pairs. The data roots are the October 5, 2026 runs `steady_purity_gap_v1` (18:41:36 UTC) and `square_steady_purity_gap_v1` (22:54:46 UTC). Exact paths, configuration and source hashes, convergence checks, and unrounded values are retained in the machine-readable validation record. Small-Fock-space checks test the reset identity, covariance convention, exterior-power spectra, operator sectors, and continuous-time examples. No new dynamics, fits, or thermodynamic extrapolations enter this consolidation.

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